



Department of Computer Science Gujarat University
M.Sc.(Artificial Intelligence & Machine Learning) – II

Axiom-I

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Presentation Roadmap

- Introduction and motivation
- Why the name Axiom-I
- Research problem, objectives, and gap
- Methodology and system architecture
- Mathematics of decision and calibration
- Implementation and diagnostics
- Results and comparative analysis
- Limitations, challenges, and future work
- Conclusion and references



Introduction

- AI-generated media has created major risks for trust, journalism, governance, and digital evidence.
- Deepfake images and videos are now temporally and visually realistic enough to fool non-experts and sometimes trained reviewers.
- A practical detector must provide both prediction and reasoning, not just a binary label.
- This work presents an explainable, full-stack multimedia forensics system focused on spatial imagery and temporal consistency.



Why the Name "Axiom-I"

- In mathematics, an axiom is a foundational truth.
- This project's mission is to mathematically recover truth from synthetic/manipulated visual content.
- "I" denotes first deployable iteration of a scalable forensic framework.
- The name also reflects a forensic "eye" that inspects hidden signals beyond human perception.
- Conceptually: from digital uncertainty to measurable, explainable truth.



Research Problem

- Existing detectors often degrade under compression, blur, lighting shift, and resizing.
- Many pipelines output labels without transparency, reducing practical trust.
- Single-cue systems are fragile against spatial and temporal manipulation diversity.
- Need: robust multi-cue analysis + explainable outputs + reliability diagnostics.
- Core question: How to design a reliable and explainable Real vs Fake multimedia (Image + Video) forensics pipeline?



Research Question and Hypothesis

- **Research Question:** Can multi-signal fusion and temporal analysis with continuous calibration outperform single-signal heuristic detection under varied media conditions?
- **Hypothesis:** Combining heterogeneous spatial and temporal forensic cues with learned calibration improves both accuracy and reliability.
- **Secondary Hypothesis:** Explainability and diagnostics improve user trust and error analysis quality.



Objectives of the Research

- Build a hybrid multimedia-forensics pipeline combining signal-processing and learned features.
- Support face-aware image analysis alongside holistic temporal video analysis.
- Provide interpretable outputs: component scores, confidence, and mathematical decision factors.
- Build end-to-end platform: frontend workspace + backend APIs.
- Add feedback-driven calibration, remote telemetry, and diagnostic monitoring.



Scope of This Thesis

- **Implemented Scope:** Full multimedia (Image + Video) Real/Fake forensics with explainable output and adaptive calibration.
- **Deployed Architecture:** Dual-environment (Local & Cloud) with live remote telemetry syncing.
- **Roadmap Scope:** Extend into multimodal forensics via topology/physics-inspired methods.



Literature Review

- FaceForensics++ defined large-scale manipulated-face benchmarks.
- MTCNN enabled robust face detection/alignment in practical pipelines.
- Vision Transformer models improved global representation quality.
- Frequency-domain and wavelet cues showed utility in synthetic artifact detection.
- Retinex/illumination theory supports physics-informed feature extraction.
- Optical flow and temporal reliability metrics are essential beyond raw spatial accuracy.



Literature Gap

- Few systems jointly combine:
- Multi-cue spatial and temporal fusion
- Explainability-first mathematical outputs
- Full-stack deployability (Local and Cloud Architecture)
- Feedback-aware reliability adaptation and remote telemetry
- Most solutions optimize raw model accuracy but under-serve transparency and trust analytics.



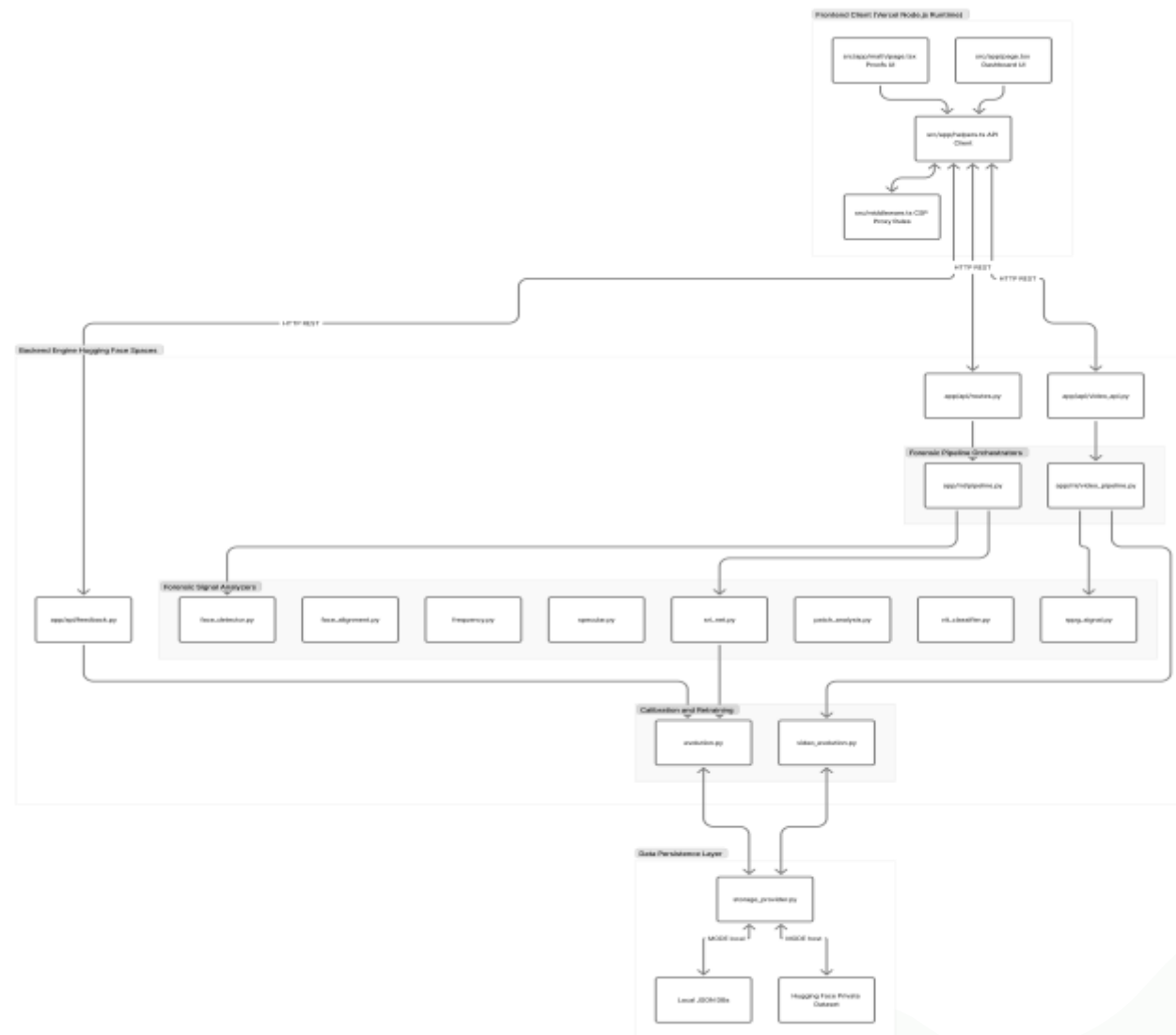
Proposed System (High-Level Architecture)

- User uploads image or video in frontend analysis workspace.
- Backend validates, decodes, and routes media into its respective forensic pipeline.
- Pipeline computes multi-signal features (spatial anomalies for images, temporal inconsistencies for videos).
- Fused heuristic score (via Weighted Average or Noisy-OR Logic) is calibrated by learned model.
- Final response includes verdict, confidence, score breakdown, and diagnostics metadata.
- User feedback updates reliability statistics and future calibration behaviour.



D1: End-to-End System Architecture

- System integrates frontend, backend, ML pipelines (Image & Video), and feedback loop into a unified dual-environment architecture.





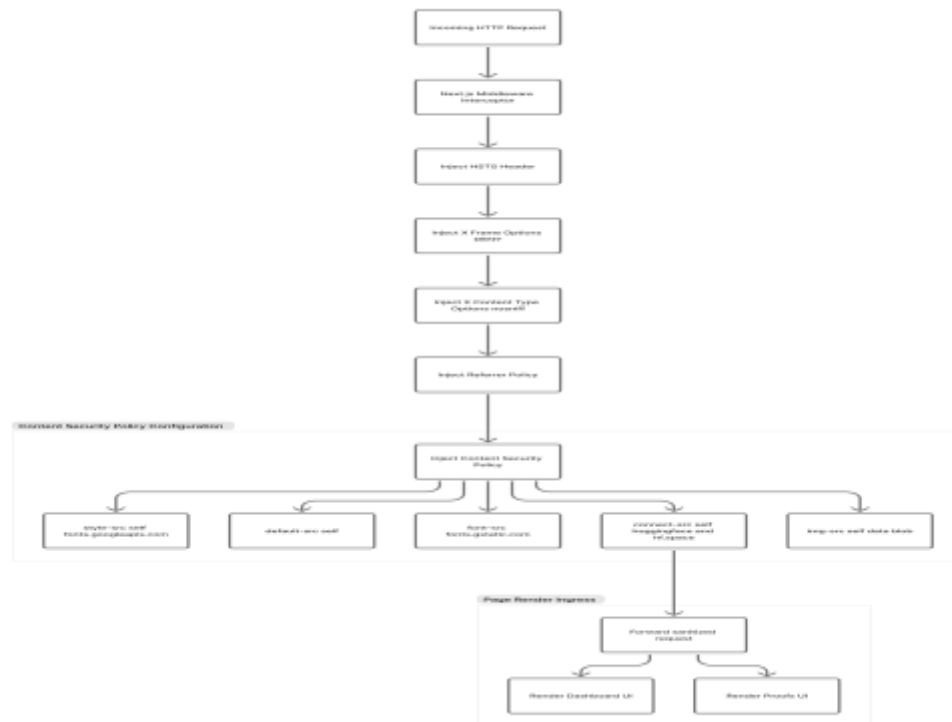
D2: Frontend User Journey Flow



- User journey has three stages: initialization, analysis, and feedback loop ensuring continuous system improvement through remote telemetry.



D3: Frontend Security Middleware Flow

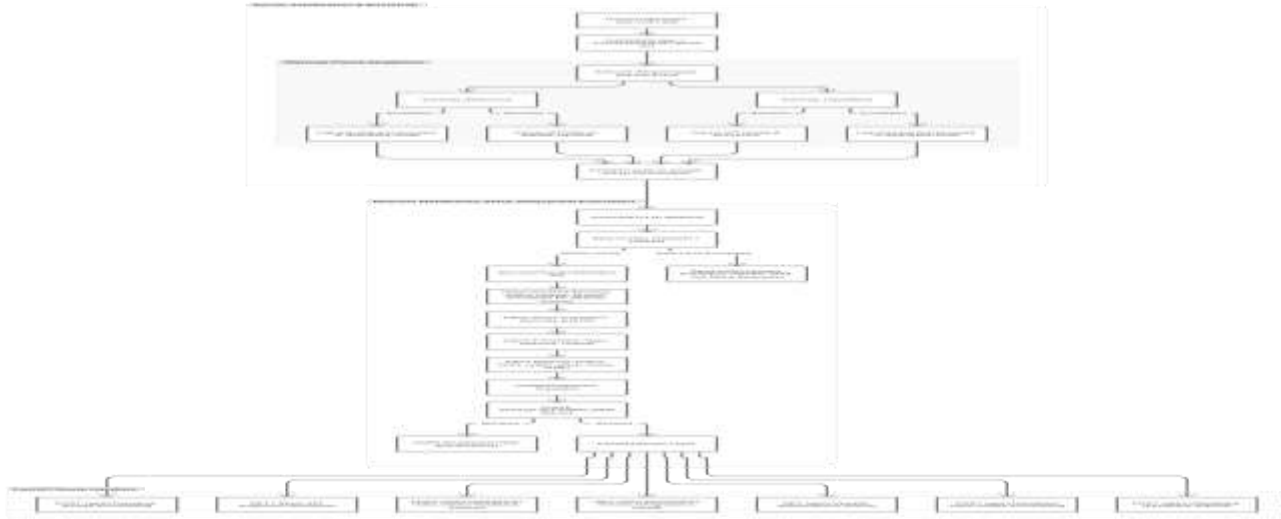


- All incoming requests are validated and secured using strict headers before reaching the backend.
- **Frontend Hosting Link:** <https://axiom.dpdns.org/>

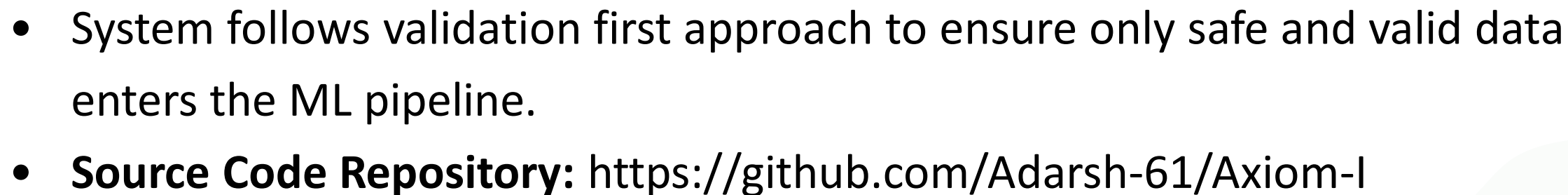


Backend Request Lifecycle

- Request validation: content type, size, decode validity, resolution safety.
- Media preprocessing and mode selection (Image pipeline or Video pipeline).
- Feature extraction from multiple modules running concurrently via thread offloading.
- Fusion and calibration.
- Structured API response generation.
- Optional feedback endpoint updates metrics, adaptive calibration state, and syncs telemetry.



- Backend initializes configuration, security layers, and routing before serving requests.
- **Backend Hosting Link:** <https://huggingface.co/spaces/Adarsh-61/axiom-backend>





Core Forensic Signals Used

- **Image:** Specular consistency, Frequency anomaly, Topology complexity
- **Image:** Patch consistency (PRNU), Wavelet anomaly, ViT-based learned cue
- **Video:** Global Scene Coherence, Optical Flow Consistency
- **Video:** Temporal Frequency Anomalies, Biometric Stability
- **Video:** Temporal Backbone (Learnable)
- **Quality features:** sharpness, blockiness, colorfulness, resolution

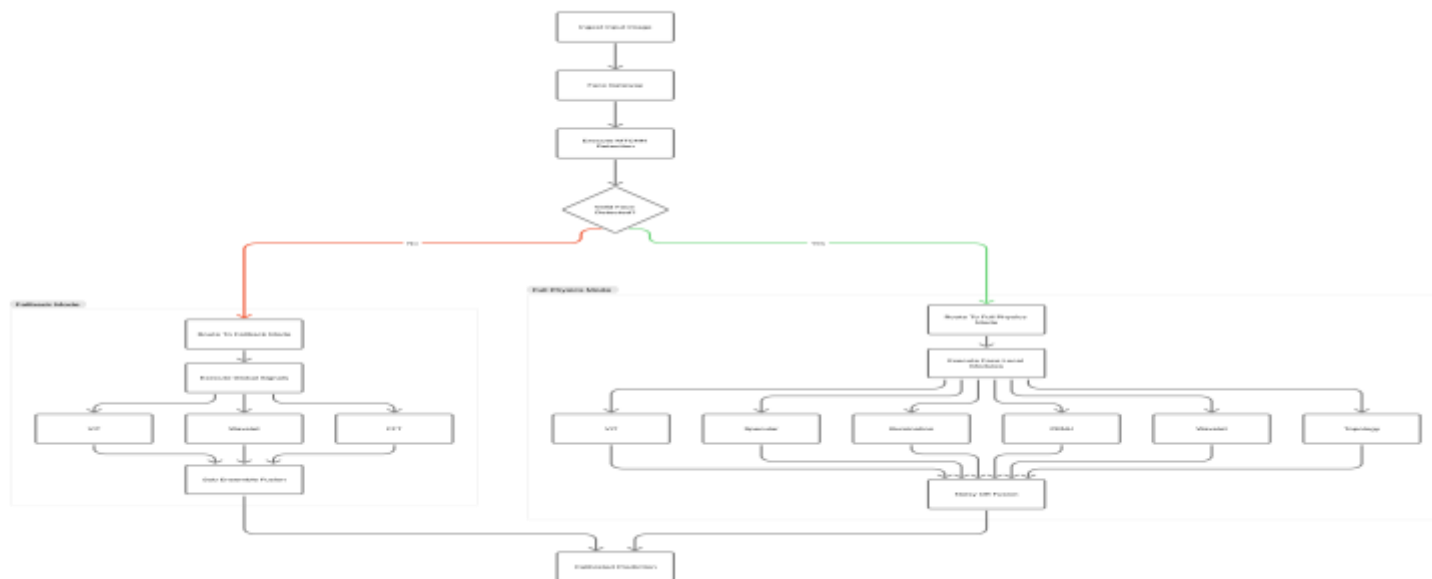


Dual Analysis Modes

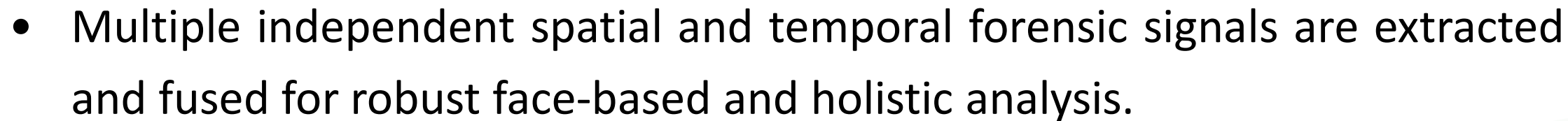
- Full Physics Mode (face detected): rich face-region forensic stack.
- Fallback Mode (no reliable face): full-image frequency + wavelet + ViT pathway.
- This avoids hard-fail behavior when face detection is weak.
- Improves robustness for diverse real-world uploads.



D6: Pipeline Mode Switch



- System dynamically selects analysis path based on whether a face is detected in the image, or routes to the Video pipeline for temporal media.



- Multiple independent spatial and temporal forensic signals are extracted and fused for robust face-based and holistic analysis.



D8: Fallback Image Analysis Path

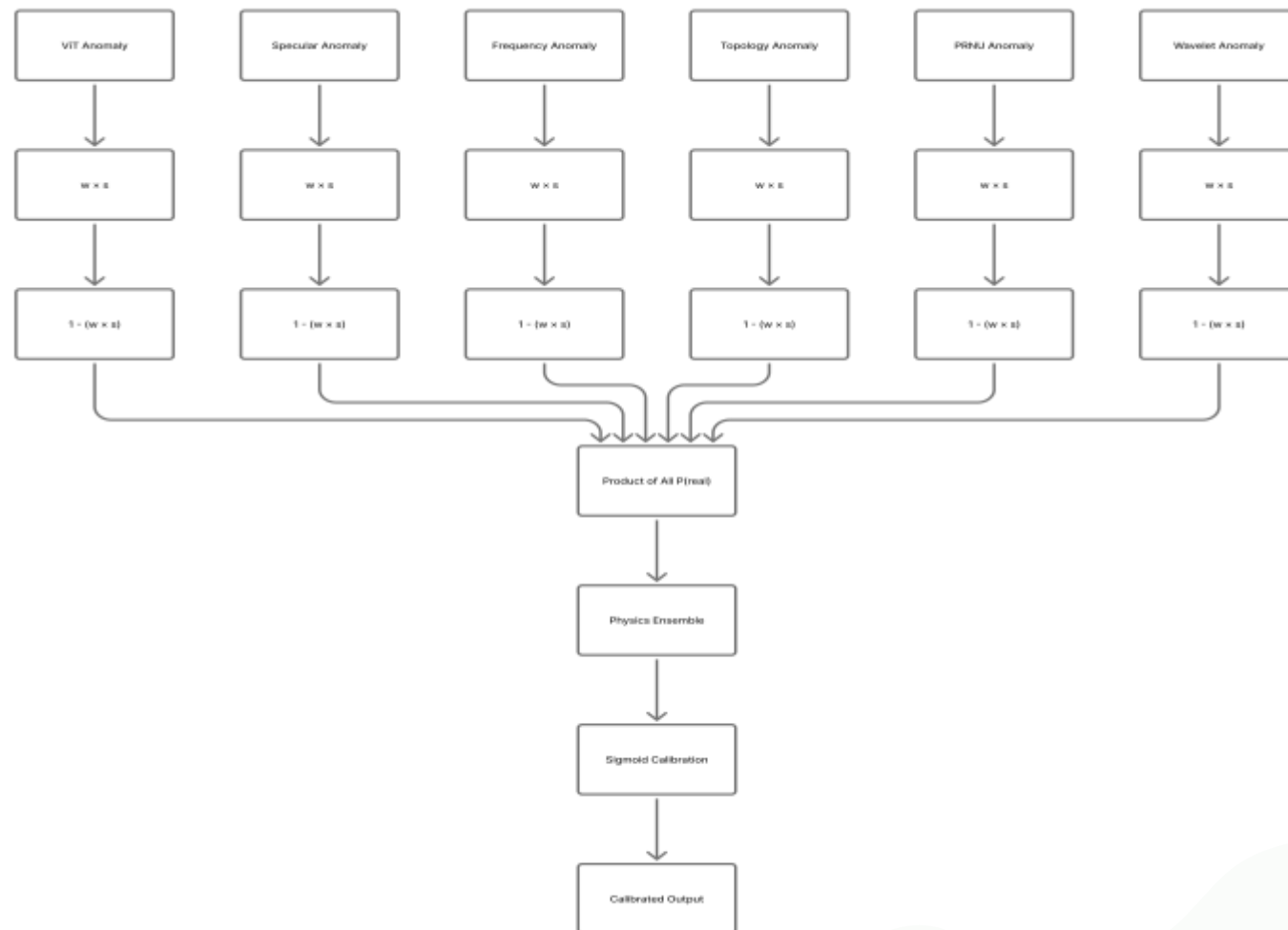


- When face is not detected, system uses global image signals to maintain reliability without blocking the pipeline.



D9: Signal Modules Fusion Flow

- Different signals capture different spatial and temporal artifacts and their combination improves overall decision robustness.





Mathematical Decision Layer

(1) Image Fusion: Weighted spatial blending -> Sigmoid Activation -> Score

(2) Video Fusion: $P(\text{Fake}) = 1 - \prod (1 - P(S_i))$ (Noisy-OR Logic)

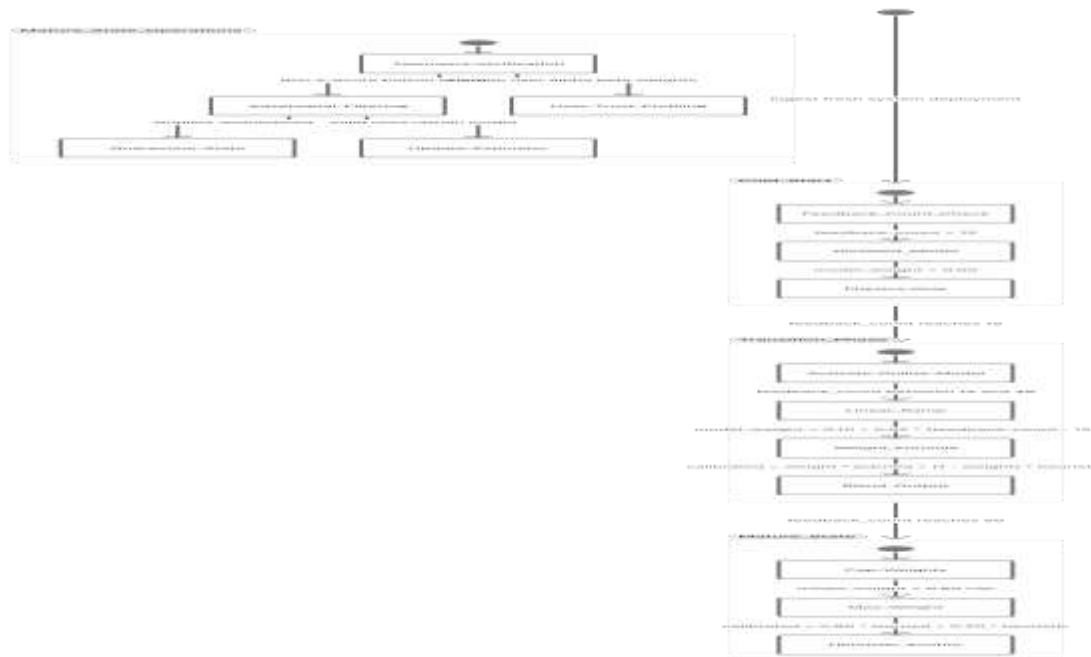
(3) $s = w \times s(\text{learned}) + (1 - w) \times s_h$, $0.60 \leq w \leq 0.90$

(4) Verdict: Fake if $s \geq 0.5$ | Real if $s < 0.5$

(5) Confidence = $2 \times |s - 0.5|$



D10: Calibration Model Lifecycle

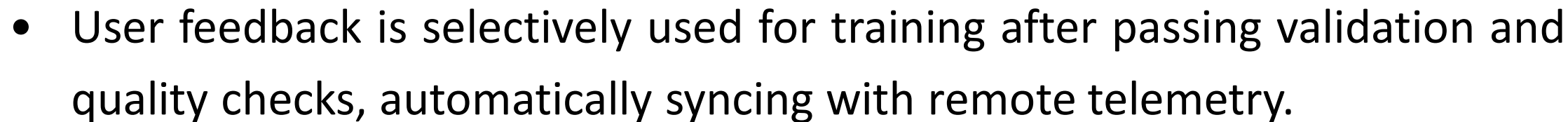


- System combines learned and heuristic scores to improve prediction accuracy, temporal stability, and overall reliability.



Calibration and Adaptive Learning

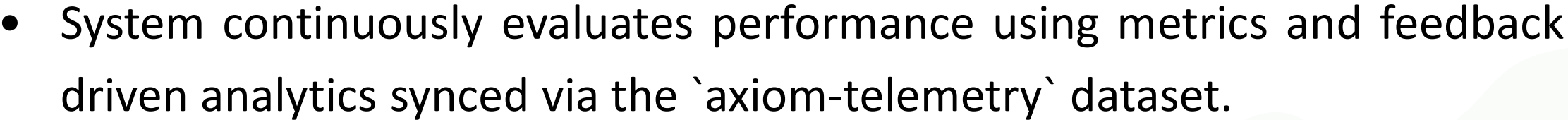
- Seed features are cached from labeled Real/Fake dataset.
- Calibration model selected by cross-validated loss among candidates.
- Best selected family in current run: HistGradientBoosting.
- User feedback can influence adaptive behavior via eligibility checks and trust-weighting.
- Reliability metrics and history are persisted via Hugging Face Datasets telemetry.





Explainability and Diagnostics

- Signal-level breakdown visible in output via UI gauges and LaTeX formulas.
- Final score decomposition: heuristic, learned score, blending weight.
- Feedback diagnostics include confusion matrices, calibration quality, and trend history.
- Exclusion reasons are tracked when feedback is not training-eligible.
- Trust summary supports responsible, auditable behavior.





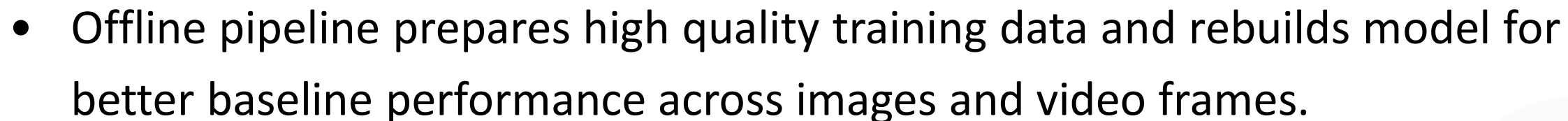
Data Collection

- Binary class organization: Real and Fake.
- Quality diversity considered: resolution, compression, illumination variation, temporal noise.
- Runtime uploads used for practical interactive validation.
- Current large training-feature pass generated from:
 - Real media: 6,290
 - Fake media: 7,000
 - Total: 13,290 samples



Experimental Setup

- Quantitative design-and-evaluation oriented methodology.
- Multi-signal extraction (spatial + temporal) followed by calibrated decision.
- Internal comparison:
 - Baseline heuristic (pre-calibration)
 - Calibrated model (cross-validated)
- Metrics tracked: Accuracy, Precision, Recall, F1, AUC, Log-loss, Brier Score.



- Offline pipeline prepares high quality training data and rebuilds model for better baseline performance across images and video frames.



Results Summary (Verified)

- Samples used in evaluation: **13,290**
- Selected calibration family: **hist_gradient_boosting**
- Heuristic accuracy: **57.10%**
- Calibrated CV accuracy: **69.72%**
- Absolute gain: **+12.62 percentage points**
- AUC improved: **0.5748 → 0.7772**
- Log-loss improved: **0.7893 → 0.5398**
- Brier score improved: **0.2778 → 0.1867**



Class-wise Performance (Fake as Positive)

- **Heuristic Baseline:**

Precision (Fake): 56.31%

Recall (Fake): 82.84%

F1 (Fake): 67.04%

- **Calibrated Model:**

Precision (Fake): 67.66%

Recall (Fake): 81.43%

F1 (Fake): 73.91%

- **Key observation:** significant precision and F1 improvement with slight recall trade-off.



Confusion Matrix Comparison

Heuristic (Baseline)

	Predicted: Real	Predicted: Fake
Actual: Real	TN = 1,790	FP = 4,500
Actual: Fake	FN = 1,201	TP = 5,799

Calibrated Model

	Predicted: Real	Predicted: Fake
Actual: Real	TN = 3,566	FP = 2,724
Actual: Fake	FN = 1,300	TP = 5,700

Key improvement: False Positives reduced by 1,776 • True Negatives nearly doubled (1,790 → 3,566)



Interpretation of Results

- Calibration reduced over-triggering on Real class.
- Better balance between detection sensitivity and practical reliability.
- Multi-cue spatial and temporal fusion + calibration is more stable than direct threshold-only scoring.
- Reliability metrics confirm improved probabilistic quality, not just raw correctness.



Traditional vs Axiom-I

- Traditional systems often rely on one dominant model stream.
- Axiom-I combines heterogeneous spatial and temporal forensic cues.
- Traditional outputs are often opaque; Axiom-I provides signal-wise explanation.
- Traditional pipelines under-emphasize reliability calibration.
- Axiom-I integrates calibration, confidence logic, and feedback diagnostics in one loop.



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Contributions to the Field

- Practical explainable Real/Fake multimedia-forensics framework.
- Multi-cue fusion (Bayesian and Weighted) with calibration-aware final decisioning.
- End-to-end deployable architecture with remote telemetry and feedback loop.
- Trust-oriented design moving beyond black-box prediction.



Limitations

- Current thesis scope is Image and Video focused.
- Generalization can vary on unseen manipulation families.
- Extreme compression/lighting can degrade some spatial and temporal cues.
- Feedback quality depends on user correctness and consistency.
- Heavy ML models require thread-safe concurrency handling in production.



Challenges Faced

- Harmonizing score scales across heterogeneous Image and Video modules.
- Achieving thread-safe concurrency for heavy ML models without blocking the backend.
- Balancing precision-recall while preserving confidence meaning.
- Handling weak/no-face cases without failing pipeline.
- Maintaining UI simplicity while exposing rich mathematical diagnostics.



Future Work (Near-Term Practical)

- Stronger external holdout benchmarking and domain-shift testing.
- Automated report export for diagnostics and calibration trends.
- Improved uncertainty estimation and threshold personalization.
- Speed optimization for real-time or edge-like deployment.
- Better active-learning workflows from verified user feedback.



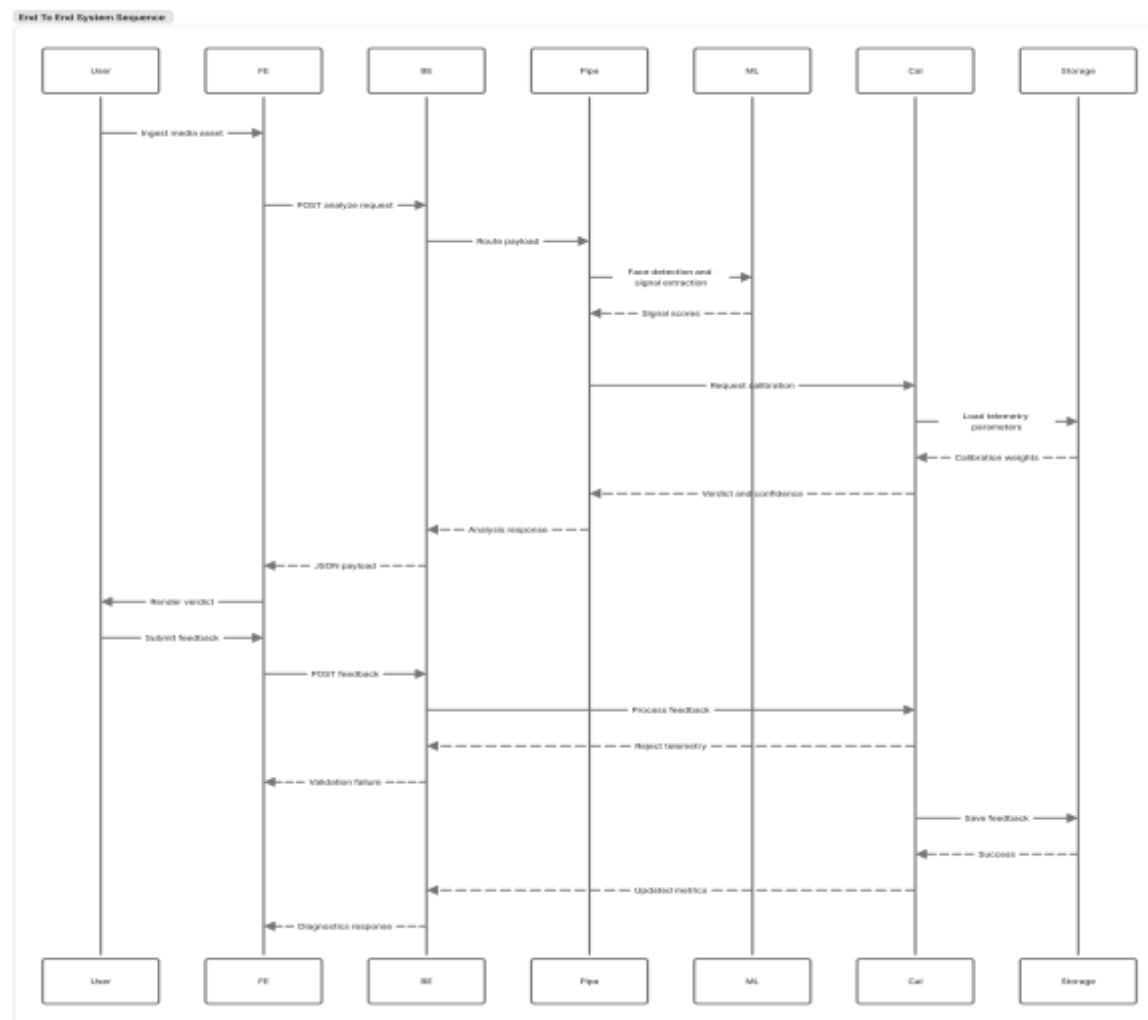
Future Work (Advanced Research Track)

- Extend to multimodal forensics: text + image + video.
- Evaluate topology-based text forensics (intrinsic dimensionality style methods).
- Explore physics-intensive image checks (advanced reflection consistency).
- Explore phase/Eulerian video biological consistency checks.
- Keep strict separation between implemented production module and exploratory research modules.



D14: End-to-End System Sequence

- System processes input, generates spatial/temporal predictions, collects feedback, and continuously improves via remote telemetry in a closed loop.





Conclusion

- Axiom-I addresses a critical digital-trust problem using explainable hybrid multimedia forensics.
- The system is not only predictive but diagnostically auditable.
- Adding the Video Pipeline and continuous remote telemetry matured Axiom-I into a production-grade platform.
- This work establishes a strong foundation for trustworthy, scalable multimedia forensics research.



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- Deepfake Detection Using Eulerian Video Magnification and Deep Learning (2024).
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Thank You!